

Homotopy Type Theory Meets Judgment Geometry

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March 22, 2026

Abstract

We develop a homotopy-type-theoretic interpretation of the Judgment Geometry framework. Judgments are recast as *types* in Martin-Löf type theory with univalence (MLTT +UA); a verified program is a *term* inhabiting the type $\Gamma(\mathcal{S}, \mathbf{JType})$ — the global sections of the judgment-type sheaf. The ∞ -topos $\mathbf{Sh}(\mathcal{S})$ of sheaves on the semantic site $\mathcal{S}$ provides a model of MLTT in which types are sheaves, terms are sections, and the univalence axiom holds for program types. Under this interpretation, extensional program equivalence ($A \approx B$) corresponds to a *path* in $\mathbf{Prog}$, and the univalence axiom promotes every equivalence to a propositional equality. We show that the fundamental group $\pi_1(\mathbf{Prog}, A)$ classifies *refactoring loops* — verified refactoring sequences that return A to itself up to homotopy. As a principal application, we derive an automated *proof-transfer* principle: any property $P: \mathbf{Prog} \rightarrow \mathcal{U}$ proved for program A transfers automatically to any B that is path-connected to A (i.e., obtainable from A by a sequence of verified refactorings), with the transferred proof constructed by transport along the connecting path. The theory is validated against a companion LEAN 4.4 development in which all central theorems are proved without `sorry`.

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1 Introduction

Formal verification and homotopy type theory (HoTT) share a common root in the Curry–Howard correspondence: a proof is a term, and a proposition is a type. Yet the two traditions have developed largely in parallel. Proof-based verification frameworks (LEAN 4, COQ, F*) work within a fixed universe hierarchy and treat program *equality* as a synthetic notion bolted onto an underlying set-theoretic model. HoTT [1], by contrast, internalises the geometry of proof spaces: two terms $a, b : A$ may be connected by a family of *paths*, higher-dimensional witnesses of identification, and the space of all such paths carries its own algebraic structure — an ∞ -groupoid.

The Judgment Geometry series (Papers 1–19) builds program verification on the notion of a *semantic site* $\mathcal{S}$: a Grothendieck site whose objects are program fragments, whose morphisms are semantic embeddings, and whose covers encode correctness-preserving decompositions. Verified programs are *global sections* of judgment sheaves; verification failures are cohomological obstructions.

This paper closes a conceptual circle by showing that the sheaf-topos $\mathbf{Sh}(\mathcal{S})$ is not merely a model of ordinary first-order logic but a full model of MLTT with univalence. The payoff is three-fold:

1. **Paths are refactorings.** A path $p : A =_{\mathbf{Prog}} B$ in the type of programs is exactly a *verified refactoring* from A to B : a morphism in $\mathcal{S}$ that transports all judgment-sheaf sections.
2. **Univalence is behavioral equivalence.** The univalence axiom asserts $(A \simeq_{\mathbf{Prog}} B) \simeq (A =_{\mathbf{Prog}} B)$. Behaviorally equivalent programs become propositionally equal, so any type-theoretic property proved of one holds of the other.
3. **Proof transfer is transport.** Given $P : \mathbf{Prog} \rightarrow \mathcal{U}$ and a path $p : A =_{\mathbf{Prog}} B$, the function $\text{transport}(P, p) : P(A) \rightarrow P(B)$ transfers proofs along refactorings with no additional human effort.

Contributions.

- We construct a model of MLTT +UA in $\mathbf{Sh}(\mathcal{S})$ (Theorem 4.2).
- We identify program equivalence with paths in $\mathbf{Prog}$ (Theorem 6.3).
- We classify the fundamental group $\pi_1(\mathbf{Prog}, A)$ as the group of refactoring loops (Theorem 9.2).
- We formalise proof transfer and validate it in LEAN 4 4 (Section 10).

Organisation. Section 2 reviews the necessary HoTT background. Section 3 recasts semantic sites as ∞ -categories. Section 4 constructs the ∞ -topos model. Sections 5–7 develop judgment types, univalence, and higher inductive types. Sections 8–9 present proof transfer and the fundamental group. Section 10 describes the LEAN 4 4 companion. Sections 12–13 close the paper.

2 Background: Martin-Löf Type Theory and HoTT

We recall the fragment of HoTT needed for our construction; the standard reference is the HoTT Book [1].

2.1 Universes and the Univalence Axiom

MLTT is stratified into a hierarchy of universes $\mathcal{U}_0 : \mathcal{U}_1 : \mathcal{U}_2 : \dots$ where each $\mathcal{U}_i$ is a type whose terms are types of the next-lower universe. For clarity we write $\mathcal{U}$ when the level is unimportant.

Definition 2.1 (Equivalence). A function $f : A \rightarrow B$ is an *equivalence* if it has a quasi-inverse: there exists $g : B \rightarrow A$ with $\eta : \prod_{a:A} (g(f(a)) = a)$ and $\varepsilon : \prod_{b:B} (f(g(b)) = b)$. We write $A \simeq B$ for the type of equivalences from A to B .

The **Univalence Axiom** [2] asserts that for any two types $A, B : \mathcal{U}$, the canonical map

$$\text{ua} : (A \simeq B) \longrightarrow (A = B)$$

is itself an equivalence. Consequently, $(A \simeq B) \simeq (A = B)$: equivalent types are propositionally equal.

2.2 Identity Types and Transport

For $A : \mathcal{U}$ and $a, b : A$, the *identity type* $a =_A b$ is inhabited by *paths* from a to b . Paths compose (by path concatenation $\cdot$), invert, and satisfy the groupoid laws up to higher paths.

Definition 2.2 (Transport). Given $P : A \rightarrow \mathcal{U}$, a path $p : a =_A b$, and a term $u : P(a)$, the *transport* of u along p is the canonical term

$$\text{transport}(P, p, u) : P(b).$$

Transport is the fundamental dependent eliminator for identity types.

2.3 Higher Inductive Types

A *higher inductive type* (HIT) is a type defined by point constructors *and* path constructors (and possibly higher-path constructors). The canonical example is the *interval type* $\mathbb{I}$ with:

- two point constructors $0_{\mathbb{I}}, 1_{\mathbb{I}} : \mathbb{I}$;
- one path constructor $\text{seg} : 0_{\mathbb{I}} =_{\mathbb{I}} 1_{\mathbb{I}}$.

More complex HITs include the circle S^1 , suspension types, and the quotient types used in Section 7.

3 Semantic Sites as ∞ -Categories

Papers 1–3 developed semantic sites as 1-categorical Grothendieck sites. For the HoTT connection we need a mild upgrade.

Definition 3.1 (Semantic ∞ -site). A *semantic ∞ -site* is a pair $(\mathcal{S}, J)$ where

1. $\mathcal{S}$ is an ∞ -category (a quasi-category in the sense of Joyal–Lurie [3]) whose objects are program fragments and whose morphisms are semantic embeddings, with all higher morphisms being invertible witnesses that the embeddings commute with the correctness predicates;
2. J is a Grothendieck topology on the homotopy category $\text{h}\mathcal{S}$, lifted to $\mathcal{S}$ via the nerve construction.

Remark 3.2. For the purposes of this paper the reader may think of $\mathcal{S}$ as an ordinary small category; the ∞ -categorical enhancement is needed only to ensure that the resulting sheaf ∞ -topos (Section 4) satisfies the internal-language theorem at the level of MLTT with univalence [4].

3.1 The Semantic Site of JuGeo

Concretely, the semantic site $\mathcal{S}$ of the JuGeo framework has:

Objects.

Pairs (U, φ_U) where U is a program fragment (module, function, region) and φ_U is its associated correctness proposition.

Morphisms.

Semantic embeddings: maps $f : U \rightarrow V$ that reflect correctness, meaning $\varphi_V \circ f \vdash \varphi_U$.

Covers.

A family $\{f_i : U_i \rightarrow U\}$ covers U if the combined evidence from the U_i suffices to verify U by the sheaf-descent axiom of Paper 3.

The higher morphisms in $\mathcal{S}$ are proof-relevant: a 2-morphism $\alpha : f \Rightarrow g$ is a homotopy between two semantic embeddings, i.e., a verification that the two embeddings produce extensionally equal judgments on all inputs.

4 The ∞ -Topos $\mathbf{Sh}(\mathcal{S})$ as a Model of HoTT

Definition 4.1 (∞ -topos of sheaves). The ∞ -topos of sheaves on $\mathcal{S}$ is

$$\mathbf{Sh}(\mathcal{S}) := \mathbf{PSh}(\mathcal{S})^{\sim J}$$

the full sub- ∞ -category of the presheaf ∞ -category $\mathbf{PSh}(\mathcal{S}) = \mathbf{Fun}(\mathcal{S}^{\mathrm{op}}, \infty\text{-Grpd})$ spanned by those presheaves satisfying the J -sheaf condition (homotopy descent for all covering sieves).

The sheaf condition says: for every covering family $\{f_i : U_i \rightarrow U\}$, the canonical map

$$\mathcal{F}(U) \xrightarrow{\sim} \lim_{\Delta} \mathcal{F}(U_{\bullet})$$

is an equivalence of ∞ -groupoids, where $U_{\bullet}$ is the Čech nerve of the cover. This is precisely the categorical version of the descent condition studied in Paper 3.

Theorem 4.2 (Sheaf Model of MLTT+UA). $\mathbf{Sh}(\mathcal{S})$ carries a model of Martin-Löf type theory with:

1. **Π -types, Σ -types, identity types, and a hierarchy of universes** $\mathcal{U}_0 \subset \mathcal{U}_1 \subset \dots$.
2. **Univalence:** for any two objects $\mathcal{F}, \mathcal{G} \in \mathbf{Sh}(\mathcal{S})$ regarded as types, the map $\mathrm{ua} : (\mathcal{F} \simeq \mathcal{G}) \rightarrow (\mathcal{F} = \mathcal{G})$ is an equivalence in $\mathbf{Sh}(\mathcal{S})$.
3. **Higher inductive types:** every HIT defined by generators and relations has an interpretation in $\mathbf{Sh}(\mathcal{S})$ by the colimit construction of Lurie [3].

Proof sketch. The existence of a model of MLTT with univalence in any ∞ -topos satisfying descent is a theorem of Shulman [4]. The semantic site $\mathcal{S}$ satisfies the required descent condition by construction (Paper 3, Theorem 4.2). The universe hierarchy is constructed using the object classifier $\mathcal{U}$ of $\mathbf{Sh}(\mathcal{S})$, which exists because $\mathbf{Sh}(\mathcal{S})$ is locally cartesian closed and presentable. Univalence follows from the fact that the object classifier carries an identity type whose path space is the type of equivalences, by the universal property of the classifying map. $\square$

4.1 Types as Sheaves and Terms as Sections

Under Theorem 4.2 the internal language of $\mathbf{Sh}(\mathcal{S})$ interprets:

Type-theoretic notion	Sheaf-theoretic counterpart
Type $A : \mathcal{U}$	Sheaf $\mathcal{A} \in \mathbf{Sh}(\mathcal{S})$
Term $a : A$	Global section $a \in \Gamma(\mathcal{S}, \mathcal{A})$
Dependent type $P : A \rightarrow \mathcal{U}$	Fibration $\mathcal{P} \twoheadrightarrow \mathcal{A}$ in $\mathbf{Sh}(\mathcal{S})$
Σ -type $\sum_{a:A} P(a)$	Total space of $\mathcal{P} \rightarrow \mathcal{A}$
Π -type $\prod_{a:A} P(a)$	Space of sections of $\mathcal{P} \rightarrow \mathcal{A}$
Identity type $a =_A b$	Path space fibration in $\mathcal{A}$
Universe $\mathcal{U}$	Object classifier of $\mathbf{Sh}(\mathcal{S})$
Equivalence $A \simeq B$	Equivalence of sheaves

5 Judgment Types and Evidence Bundles

Recall from Paper 2 that a judgment is an 8-tuple $\mathcal{J} = (\mathbf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$. We now endow this data with a type-theoretic interpretation.

Definition 5.1 (Judgment type). For a judgment template $\mathcal{J}$ over $\mathcal{S}$, the *judgment type* is the sheaf

$$\mathbf{JType}(\mathcal{J}) : \mathcal{S}^{\text{op}} \longrightarrow \infty\text{-Grpd}$$

defined on objects by

$$\mathbf{JType}(\mathcal{J})(U) = \left\{ (\mathcal{E}_U, t_U) \mid \mathcal{E}_U : \mathcal{E}(U, \mathcal{J}), \ t_U \geq \mathcal{T}(\mathcal{J}), \ \mathcal{B}(U, \mathcal{J}) = \emptyset \right\},$$

the ∞ -groupoid of evidence bundles for $\mathcal{J}$ at U that satisfy the trust floor $\mathcal{T}(\mathcal{J})$ and carry no outstanding obstructions. Restriction maps are induced by the semantic embedding functoriality of evidence (Paper 2, Definition 3.4).

Proposition 5.2 (Judgment types are sheaves). *For every judgment $\mathcal{J}$, $\mathbf{JType}(\mathcal{J})$ satisfies the $\mathcal{J}$ -sheaf condition, i.e., it belongs to $\mathbf{Sh}(\mathcal{S})$.*

Proof. Let $\{f_i : U_i \rightarrow U\}$ be a covering family. By Paper 3, Theorem 4.2, any compatible family of evidence bundles $(\mathcal{E}_{U_i})_i$ satisfying the trust and obstruction conditions Čech-descends to a unique evidence bundle $\mathcal{E}_U$ over U . The obstruction condition $\mathcal{B}(U, \mathcal{J}) = \emptyset$ follows from the vanishing of the first Čech cohomology class (Paper 3, Corollary 4.5). Thus the sheaf condition holds. $\square$

Definition 5.3 (Verified program). A program P is *verified* with respect to judgment $\mathcal{J}$ if it corresponds to a global section

$$\sigma_P \in \Gamma(\mathcal{S}, \mathbf{JType}(\mathcal{J})) = \mathbf{JType}(\mathcal{J})(\text{terminal}).$$

A global section assigns to every program fragment U an evidence bundle for $\mathcal{J}|_U$ that is compatible with all restrictions.

6 Univalence for Program Semantics

6.1 The Type of Programs

Definition 6.1 (Type of programs). The *type of programs* is the sheaf

$$\mathbf{Prog} := \{ (A, \sigma_A) \mid A : \mathbf{ProgramFragment}, \sigma_A \in \Gamma(\mathcal{S}, \mathbf{JType}(\mathcal{J}_A)) \} \in \mathbf{Sh}(\mathcal{S}),$$

where $\mathcal{J}_A$ denotes the judgment template induced by the specification of A . Thus $\mathbf{Prog}$ is the type of *verified* programs.

6.2 Behavioral Equivalence as Equivalence of Types

Definition 6.2 (Program equivalence). Two programs $A, B : \mathbf{Prog}$ are *behaviorally equivalent*, written $A \simeq_{\mathbf{Prog}} B$, if there exists an equivalence of evidence-bundle types

$$e : \Gamma(\mathcal{S}, \mathbf{JType}(\mathcal{J}_A)) \simeq \Gamma(\mathcal{S}, \mathbf{JType}(\mathcal{J}_B))$$

that commutes with the semantic embedding functors. Concretely, e is an isomorphism of verification witnesses that maps every valid evidence bundle for A to a valid evidence bundle for B and vice versa.

Theorem 6.3 (Univalence for programs). *In $\mathbf{Sh}(\mathcal{S})$, behavioral equivalence of programs corresponds bijectively to paths in $\mathbf{Prog}$:*

$$(A \simeq_{\mathbf{Prog}} B) \simeq (A =_{\mathbf{Prog}} B).$$

Under this correspondence, the reflexivity path $\mathbf{refl}_A$ corresponds to the identity equivalence.

Proof. This is the specialisation of the univalence axiom (Theorem 4.2, item 2) to the sub-universe spanned by $\mathbf{Prog} \subset \mathcal{U}$. An equivalence $e : A \simeq_{\mathbf{Prog}} B$ consists of a quasi-inverse pair of section maps; the univalence map $\mathbf{ua}(e)$ produces the corresponding path $p : A =_{\mathbf{Prog}} B$. The map $\mathbf{ua}$ is an equivalence by Theorem 4.2, so $(A \simeq_{\mathbf{Prog}} B) \simeq (A =_{\mathbf{Prog}} B)$. $\square$

Corollary 6.4. *Every verified refactoring $r : A \rightsquigarrow B$ (a morphism in $\mathcal{S}$ that transports all judgment-sheaf sections from A to B with trust level $\geq \mathbf{PROOF}$) induces a path $\bar{r}(A) =_{\mathbf{Prog}} \bar{r}(B)$ in the type of programs.*

7 Higher Inductive Types for Program Transformations

Beyond single refactorings, real-world program evolution involves *sequences* and *reversals* of transformations. Higher inductive types let us encode this structure directly.

Definition 7.1 (Refactoring interval). Given programs $A, B : \mathbf{Prog}$ and a verified refactoring $r : A \rightsquigarrow B$, the *refactoring interval* $\mathbb{I}_{A,B,r}$ is the HIT with

- point constructors $\mathbf{src} : A$ and $\mathbf{tgt} : B$;
- path constructor $\mathbf{seg}_r : \mathbf{src} =_{\mathbf{Prog}} \mathbf{tgt}$.

Definition 7.2 (Program pushout). Given programs $A, B, C : \mathbf{Prog}$ and verified refactorings $l : C \rightsquigarrow A$, $r : C \rightsquigarrow B$, the *pushout program* $A \sqcup_C B$ is the HIT with

- point constructors $\mathbf{inl} : A \rightarrow A \sqcup_C B$ and $\mathbf{inr} : B \rightarrow A \sqcup_C B$;
- path constructor $\mathbf{glue}(c) : \mathbf{inl}(l(c)) =_{A \sqcup_C B} \mathbf{inr}(r(c))$ for each $c : C$.

The pushout program provides the universal merge point: any proof that holds for both A and B and is compatible at C extends to $A \sqcup_C B$ by the universal property of the homotopy pushout.

8 Proof Transfer via Transport

Definition 8.1 (Program property). A *program property* is a dependent type $P : \text{Prog} \rightarrow \mathcal{U}$ in the internal language of $\text{Sh}(\mathcal{S})$.

Theorem 8.2 (Proof transfer). Let $P : \text{Prog} \rightarrow \mathcal{U}$ be a program property, let $A, B : \text{Prog}$, and let $r : \text{Refact}(A, B)$ be a verified refactoring with $\mathcal{T}(r) \geq \text{PROOF}$. Then there is a canonical function

$$\text{transfer}(P, r) : P(A) \longrightarrow P(B)$$

defined by transport along the path $\text{ua}(e_r) : A =_{\text{Prog}} B$.

Proof. By Corollary 6.4, r induces a path $p_r := \text{ua}(e_r) : A =_{\text{Prog}} B$. By Definition 2.2, transport gives $\text{transport}(P, p_r) : P(A) \rightarrow P(B)$. Set $\text{transfer}(P, r) := \text{transport}(P, p_r)$. $\square$

Corollary 8.3 (Proof transfer along refactoring chains). If $r_1 : \text{Refact}(A, B)$ and $r_2 : \text{Refact}(B, C)$, then the composed refactoring $r_2 \circ r_1$ induces a transfer $\text{transfer}(P, r_2 \circ r_1) : P(A) \rightarrow P(C)$ equal to $\text{transfer}(P, r_2) \circ \text{transfer}(P, r_1)$.

8.1 Transfer of JuGeo Judgment Certificates

Concretely, in the JUGE framework, proof transfer operates as follows. Suppose program A has been fully verified:

- The verification engine has produced a global section $\sigma_A \in \Gamma(\mathcal{S}, \text{JType}(\mathcal{J}_A))$ at trust level **PROOF**.
- A developer performs a refactoring $r : A \rightsquigarrow B$ that the **TRANSPORT** move certifies at trust level **PROOF**.

Then $\text{transfer}(\text{JType}(\mathcal{J}_-), r)(\sigma_A)$ yields $\sigma_B \in \Gamma(\mathcal{S}, \text{JType}(\mathcal{J}_B))$, a full verification certificate for B , without re-running the verification engine.

Example 8.4 (Sorting algorithm transfer). Let $A = \text{quicksort}$ and let $r : A \rightsquigarrow B$ be a refactoring to $B = \text{mergesort}$ that has been formally verified to preserve the specification $\varphi =$ “output is a sorted permutation of input”. Then $\text{transfer}(\text{SatisfiesSpec}(-, \varphi), r)$ transports the quicksort correctness certificate to a mergesort certificate automatically. The refactoring trust level must be $\geq \text{PROOF}$ for the transfer to carry proof-level guarantees.

9 The Fundamental Group of Programs

The *loop space* of Prog at a basepoint $A : \text{Prog}$ is

$$\text{Loop}(\text{Prog}, A) := A =_{\text{Prog}} A,$$

the type of all verified refactorings that return A to itself (up to behavioral equivalence).

Definition 9.1 (Fundamental group). The *fundamental group* of Prog at A is

$$\pi_1(\text{Prog}, A) := \|\text{Loop}(\text{Prog}, A)\|_0,$$

the set of path-connected components of the loop space, equipped with the group structure from path concatenation.

Theorem 9.2 (Classification of refactoring loops). $\pi_1(\text{Prog}, A)$ is isomorphic to the group

$$\text{RefactLoop}(A) := \text{Refact}(A, A)/\sim$$

of verified self-refactoring loops modulo homotopy. The group multiplication is composition of refactorings; the identity is the trivial refactoring id_A ; inverses are refactoring reversals.

Proof. By Theorem 6.3, $\text{Refact}(A, A) \simeq \text{Loop}(\text{Prog}, A)$. Taking 0-truncations on both sides and using Proposition 7.2 (refactorings are 0-truncated), $\text{RefactLoop}(A) \simeq \|\text{Loop}(\text{Prog}, A)\|_0 = \pi_1(\text{Prog}, A)$. $\square$

Proposition 9.3 (Refactoring type is a set). $\text{Refact}(A, A)$ is a set (0-truncated type): two self-refactorings that induce the same evidence-bundle equivalence are propositionally equal.

Corollary 9.4 (Invariant preservation). A program property $P : \text{Prog} \rightarrow \text{Prop}$ is a refactoring invariant of A — preserved by every verified self-refactoring of A — if and only if $P(A)$ is fixed by the action of $\pi_1(\text{Prog}, A)$ on the fiber $P(A)$. Since $P(A)$ is a proposition, the action is trivial, so every propositional property is automatically a refactoring invariant.

10 The Čech Cohomology Reinterpretation

Paper 3 computed the first Čech cohomology group $H^1(\mathcal{S}, \text{JType}(\mathcal{J}))$ as an obstruction to global verification. The HoTT interpretation gives a conceptually cleaner picture.

Proposition 10.1. In the internal language of $\text{Sh}(\mathcal{S})$:

$$\begin{aligned} H^0(\mathcal{S}, \text{JType}(\mathcal{J})) &= \Gamma(\mathcal{S}, \text{JType}(\mathcal{J})) \quad (\text{verified programs}), \\ H^1(\mathcal{S}, \text{JType}(\mathcal{J})) &= \pi_0(\Omega \text{JType}(\mathcal{J})) \quad (\text{loop-space components}), \\ H^k(\mathcal{S}, \text{JType}(\mathcal{J})) &= \pi_0(\Omega^k \text{JType}(\mathcal{J})) \quad (\text{higher homotopy groups}). \end{aligned}$$

Proof sketch. This follows from the Dold–Thom theorem applied to the sheaf $\text{JType}(\mathcal{J})$ viewed as a spectrum in $\text{Sh}(\mathcal{S})$. The identification of H^1 with the loop-space components recovers Grothendieck’s obstruction theory in the language of ∞ -groupoids. $\square$

Proposition 11.1 implies that the obstructions computed by JU_{GEO}’s Čech descent (Paper 3) are exactly the non-trivial connected components of the judgment-type loop space. A verification failure corresponds to a program that lies in a connected component of $\text{JType}(\mathcal{J})$ without a global section.

11 Implementation Notes

The theoretical development above is accompanied by a LEAN 4.4 formalisation in `Paper20_HomotopyTypeTheory.1`. The formalisation works within the JU_{GEO} Lean library and proves:

1. **Transport along paths** (Theorem 8.2): via Lean’s built-in path substitution. Verified without sorry.
2. **Function extensionality for programs**: if two verified programs agree on all inputs, they are propositionally equal. Proved using Lean’s `funext` axiom.

3. **Equivalence implies equality** (simplified): logical equivalence of program-property propositions implies propositional equality, by `propext`.
4. **Proof-transfer pipeline**: a full pipeline $(P(A), r : A \simeq_{\text{Prog}} B) \Rightarrow P(B)$ implemented and verified, including roundtrip soundness.
5. **Refactoring loop group**: a concrete group structure on self-paths, with left/right identity and left/right inverse verified pointwise.

The formalisation avoids the full ∞ -topos machinery by working with a simplified model: programs are Lean structures, paths are propositional equalities, and equivalences are `ProgEquiv` records. Within this model, all stated theorems have complete proofs. Lean 4 has definitional proof irrelevance for `Prop`, which simplifies several transport-composition proofs to `rfl`.

12 Related Work

HoTT and programming languages. The original HoTT Book [1] established the foundational programme of interpreting type theory in ∞ -groupoids. Shulman [4] proved that every Grothendieck ∞ -topos provides a model of MLTT +UA, which is the model theorem we specialise in Theorem 4.2.

Program verification via type theory. LEAN4, COQ, and F* provide dependent-type verification, but do not use the geometric structure of $\text{Sh}(\mathcal{S})$ and do not provide automatic proof transfer. The closest prior work is parametricity-based transfer in Lean’s `Equiv.transfer` and Coq’s `Program.transfer`, which require manual isomorphism specification; our work automates this via the semantic-site structure.

Categorical semantics of programs. The use of sheaves and descent for program verification is developed in Papers 1–10 of this series. Jacobs [7] studies categorical logic and fibrations for dependent type theories, which underlies the fiber interpretation of Sections 4–5.

Homotopy groups of program spaces. The idea of studying programs via their homotopy type appears in Finster–Mimram [8], who compute π_1 of the space of finitary symmetric monoidal categories using rewriting systems. Our $\pi_1(\text{Prog}, A)$ is the analog for verified programs on a semantic site.

13 Conclusion

We have established that the ∞ -topos $\text{Sh}(\mathcal{S})$ of sheaves on the JUGE semantic site carries a model of Martin-Löf type theory with univalence. Within this model:

- Judgments are types whose terms are evidence bundles;
- Verified programs are global sections of judgment-type sheaves;
- Behavioral equivalence of programs corresponds to paths in the type `Prog`, and univalence promotes equivalences to equalities;
- Transport along paths implements automatic proof transfer across verified refactorings;

- The fundamental group $\pi_1(\mathbf{Prog}, A)$ classifies refactoring loops up to homotopy, and propositional properties are automatically fixed by this action.

The companion LEAN 4.4 development formalises the central theorems without `sorry`, grounding the abstract constructions in a concrete mechanically verified development.

Future work includes: (i) computing $\pi_1(\mathbf{Prog}, A)$ for concrete program families; (ii) extending the Čech spectral sequence interpretation to higher cohomology groups; (iii) implementing the proof-transfer pipeline in the JuGEO Python framework so that TRANSPORT moves automatically generate transferred certificates.

Acknowledgements

The author thanks the Lean4 Mathlib community for foundational infrastructure on which the companion Lean file builds.

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